

MANISH KUMAR SINGH

Game Programmer | Unreal Engine (C++) & Unity (C#)

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SUMMARY

Systems-focused Game Programmer with hands-on experience architecting modular gameplay frameworks, C++ architecture, and memory optimization across Unreal Engine 5 and Unity. Proven track record in developing dynamic AI behaviors, profiling rendering pipelines for stable frame rates, and integrating middleware like Wwise. Adept at cross-functional collaboration with art and design teams to ship optimized title builds.

EXPERIENCE

Unreal Engine Developer (Intern)

Dec 2024 - Jul 2025

Secernate Games Pvt Ltd - Hyderabad, India

- › Architected modular C++ combat components and itemization systems, enabling designers to rapidly iterate on weapon mechanics.
- › Assisted in profiling rendering pipelines using RenderDoc & Unreal Insights, identifying draw call bottlenecks to support 60 FPS targets on modern consoles.
- › Built custom Behavior Tree nodes and EQS queries for enemy NPCs, improving tactical flanking and cover utilization.
- › Collaborated with backend engineers to implement serializable local and cloud-save state management.

Unity Developer

Apr 2024 - Nov 2024

Level 2 Games - Remote

- › Optimized memory footprint using Unity Addressables, reducing dynamic bundle loading times from 8s to under 2.5s.
- › Profiled asset pipelines using Unity Memory Profiler, mitigating low-memory crashes on target mobile and low-end hardware.
- › Structured scalable codebase architectures utilizing Observer and State design patterns following SOLID principles.
- › Engineered adaptive AI combat states and hosted internal playtest passes to refine player movement feel and controls.

PROJECTS

Monster Hunt

Unreal Engine 5 • C++ • Systems Architecture

- › Designed an extensible C++ inventory and weapon capability framework supporting 15+ modular skill combinations.
- › Implemented asynchronous pathfinding routines and custom EQS contexts to manage group navigation for 20+ active AI enemies.

City Bus Simulator

Unity • C# • Gameplay Systems

- › A 3D Bus Simulator built in Unity for Android and PC, featuring realistic driving mechanics, interactive vehicle systems, and immersive passenger management.
- › Optimized with custom algorithms and clean code architecture, it delivers smooth performance and a polished gameplay experience.

TECHNICAL SKILLS

Engines: Unreal Engine 5 (C++, Blueprints), Unity (C#)

Languages: C++, C#, HLSL / GLSL (Foundational)

Systems & AI: Behavior Trees, EQS, A* Pathfinding, Addressables, Memory Profiling

Tools & Tech: Wwise, RenderDoc, Unreal Insights, Git, Visual Studio

EDUCATION

B.Sc. in Computer Science & Game Development

2021 - 2025

Backstage Pass Institute of Gaming & Technology, Hyderabad

CERTIFICATIONS

- Advanced Unreal C++ - Epic Games Learning Platform
- C++ Technical Certification - CodinGame
- Unity Certified Developer